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Addressing Population Heterogeneity for HIV Incidence Estimation Based on Recency Test

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Abstract

Cross-sectional HIV incidence estimation leverages recency test results to determine the HIV incidence of a population of interest, where recency test uses biomarker profiles to infer whether an HIV-positive individual was “recently” infected. This approach possesses an obvious advantage over the conventional cohort follow-up method since it avoids longitudinal follow-up and repeated HIV testing. In this manuscript, we consider the extension of cross-sectional incidence estimation to estimate the incidence of a different target population, addressing potential population heterogeneity. We propose a general framework that incorporates two scenarios: one when the target population is a subset of the population with cross-sectional recency testing data and the other with an external target population. In addition, we propose estimators to incorporate HIV subtypes, a special type of covariate that modifies the properties of recency tests, into our framework. Through simulation studies and a data application, we demonstrate the performance of the proposed methods. We conclude with a discussion on sensitivity analysis and future work to improve our framework.

Keywords

HIV; HIV incidence; HIV subtypes; population heterogeneity; recency test

1 | Introduction

Recent years have witnessed the broad utility of cross-sectional incidence estimation, that is, estimating HIV incidence utilizing data from a cross-sectional sample. Such an approach leverages recency test results from an algorithm with single or multiple biomarkers to classify whether an HIV-positive individual was infected “recently,” for example, infected within the past year. Cross-sectional incidence estimators [1, 2] use a representative

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Conflicts of Interest

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cross-sectional sample to estimate the incidence in a target population, under assumptions regarding the epidemic in the target population and properties of the recency test algorithm [3]. This approach does not require longitudinal follow-up and repeated HIV testing, which confers it an obvious advantage over the traditional cohort follow-up approach.

Cross-sectional incidence estimation has also been proposed to serve as a “counterfactual placebo” in clinical trials where the assignment to placebo arms is not possible for ethical or operational reasons. One potential use is to evaluate the prevention efficacy of an experimental pre-exposure prophylaxis (PrEP) product in an HIV prevention trial [4], for example, in the recently implemented Gilead PURPOSE-1 [5] and PURPOSE-2 [6] trials. At the screening stage of the trial, individuals are tested for HIV status by serology, and HIV-seropositive individuals are then tested by a recency test. A subset of eligible HIV-negative individuals who consent will be enrolled in the active-arm trial, provided with the study product, and followed up. The efficacy of the investigative product is then assessed by comparing the observed incidence in the active arm and the estimated cross-sectional incidence in the screening population, under the assumption that the screening and trial populations are similar. However, such an assumption is unlikely to hold in practice. For example, inclusion and exclusion criteria may lead to differential distributions of HIV risk factors between the screening and trial populations. Directly applying the existing approach fails to account for such population heterogeneity and may lead to biased results.

Accounting for population heterogeneity in the active-arm trial design with cross-sectional incidence estimation is in line with the framework of generalizability and transportability in the statistical literature, which aims to extend causal knowledge from randomized or observational data of a study population to a different target population [7]. Many statistical methods have been proposed in the field of generalizability and transportability, including matching [8, 9], weighting-based methods [10–14], outcome regressions [10, 13–15], and combined propensity score and outcome regression methods [13, 14, 16–18]. They usually utilize observed outcomes in the study population to infer some quantities related to the same (unobserved) outcomes in the target population, leveraging similarity between the populations. However, those approaches cannot be directly applied to our setting: it is unclear how to define “outcomes” in our setting with cross-sectional HIV incidence estimation. In addition, in our setting, the “study population” includes both HIV-positive and -negative individuals, while the “target population” includes only HIV-negative individuals. It is not obvious how to establish the “similarity” of the two populations.

In this paper, we will develop a framework to address population heterogeneity for HIV incidence estimation based on the recency test. We consider two scenarios: one when a target population is a subset of the population with cross-sectional recency testing data, for example, leveraging recency testing data from screening in an active-arm trial design; and the other with an external target population. The latter scenario includes the application of incidence estimation for a target population, making use of historical recency testing data collected in the same geographic area. Such a background incidence estimate will facilitate the design of a trial to be conducted in the target population. We also propose estimators to incorporate HIV subtypes, a special covariate that modifies the properties of recency tests, into our framework. The rest of the paper is organized as follows: we describe our

framework and proposed estimators in Section 2. In Section 3, we evaluate the performance of the proposed approach through simulation studies. Section 4 describes an analysis of data from the Sabes study, a treatment-as-prevention intervention study in Lima, Peru. We conclude with a discussion in Section 5.

2 | Method

2.1 | Cross-Sectional Incidence Estimation

Two frequently used estimators for cross-sectional incidence estimation are the snapshot estimator by Kaplan and Brookmeyer [1] and an adjusted estimator by Kassanjee et al. [2]. Here, we focus on the adjusted estimator since it includes the snapshot estimator as a special case. Let T^* denote a cutoff time such that a “recent” infection is defined as an infection developed less than T^* time units ago. The incidence estimator adopts two parameters that characterize the recency test based on this cutoff: mean duration of infection (MDRI) Ω_{T^*} and false-recent rate (FRR) β_{T^*} . MDRI is the average duration of infection among those infected less than T^* time units ago and classified as test-recent; FRR is the probability of recency-test-positive for a randomly selected individual infected more than T^* time units ago. For individual $i = 1, \dots, N$ in the cross-sectional sample, let D_i be the indicator for HIV status with $D_i = 1$ indicating HIV-positive and R_i be the indicator of whether individual i tests positive for the recency test (test-recent). The cross-sectional estimator for HIV incidence by Kassanjee et al. [2] is given by

$$\hat{\lambda} = \frac{\sum_{i=1}^N D_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)} \quad (1)$$

where $\hat{\Omega}_{T^*}$ and $\hat{\beta}_{T^*}$ are estimates for MDRI and FRR from external studies with individuals with known infection time. It implicitly assumes that the HIV incidence (and prevalence) is homogeneous among individuals, and the cross-sectional estimator concerns such homogeneous incidence. In our paper, we consider scenarios where prevalence and incidence may be heterogeneous among individuals. Specifically, we assume that HIV prevalence and incidence are potentially modified by a collection of covariates $\mathbf{X}$.¹ Additionally, we build our framework upon Gao and Bannick [3], which rigorously studied the properties of this cross-sectional estimator and established further assumptions needed for the estimator to be consistent for the HIV incidence in a target population.

2.2 | Extension of Cross-Sectional Incidence Estimate Accounting for Population Heterogeneity

2.2.1 | Property of Cross-Sectional Incidence Estimator Under Population Heterogeneity: Let T be the calendar time of HIV infection for an individual and

¹We assume that $\mathbf{X}$ captures all the factors that modify HIV prevalence and incidence. In other words, there is no unmeasured confounder.

let $A(t)$ be an indicator of eligibility at time t , that is, whether an individual would be eligible to be included in the cross-sectional survey at time t . Eligibility may depend on factors such as the inclusion criteria for a clinical trial and, naturally, the individual being alive. At any calendar time t , let the prevalence and incidence of an individual in the eligible population with covariates $\mathbf{X} = \mathbf{x}$ be $p(t | \mathbf{x}) = \Pr(T \leq t | \mathbf{X} = \mathbf{x}, A(t) = 1)$ and $\lambda(t | \mathbf{x}) = \lim_{dt \rightarrow 0} \frac{1}{dt} \Pr(t \leq T < t + dt | T \geq t, \mathbf{X} = \mathbf{x}, A(t) = 1)$, respectively. We will slightly abuse the notation and write $\lambda(t | \mathbf{x}) = \Pr(T = t | T \geq t, \mathbf{X} = \mathbf{x}, A(t) = 1)$. Similar to Gao and Bannick [3], we assume that $p(t | \mathbf{x})$ and $\lambda(t | \mathbf{x})$ are constant over time in a T^* time units period prior to the cross-sectional sample and write $p(\mathbf{x})$ and $\lambda(\mathbf{x})$ for brevity. Without loss of generality, we assume that covariates $\mathbf{X}$ are continuous and let $g(\mathbf{x})$ be the probability density function of $\mathbf{X}$ in the population, where the cross-sectional sample is from.

At time t when the recency test is conducted, we let the test-recent probability of an eligible individual who has been infected for u time units be $\phi(u, t) = \Pr(R = 1 | T = t - u, A(t) = 1)$ with $u \geq 0$. Note that this form of $\phi(u, t)$ assumes that the properties of the recency test are homogeneous across individuals in the cross-sectional sample, which implies $\Pr(R = 1 | T = t - u, \mathbf{X} = \mathbf{x}, A(t) = 1) = \Pr(R = 1 | T = t - u, A(t) = 1) = \phi(u, t)$. We will delay the discussion on the incorporation of recency test modifiers, for example, HIV subtypes, to Section 2.3. Analogous to Gao and Bannick [3], we assume the calendar time when the recency test is applied is unrelated to test accuracy given a known infection duration, and we write $\phi(u) := \phi(u, t)$. Following Gao and Bannick [3], we also assume $\phi(u)$ is constant for any $u \geq T^*$. Then, by the definition of MDRI and FRR, $\Omega_{T^*} = \int_0^{T^*} \phi(u) du$ and $\beta_{T^*} = \phi(u)$ for any $u \geq T^*$. That is, MDRI is a truncated integration of the (conditional) test-recent probability (given eligible at calendar time t) from 0 to T^* , and FRR is equal to the conditional test-recent probability if the individual was infected more than T^* time units ago.

Let t be the calendar time of the cross-sectional sample. Then,

$$\begin{aligned}
 & \Pr(D = 1, R = 1 | \mathbf{X} = \mathbf{x}, A(t) = 1) \\
 &= \int_0^\infty \Pr(R = 1 | T = t - u, \mathbf{X} = \mathbf{x}, A(t) = 1) \\
 & \quad \Pr(T = t - u | \mathbf{X} = \mathbf{x}, A(t) = 1) du \\
 &= \int_0^\infty \Pr(R = 1 | T = t - u, \mathbf{X} = \mathbf{x}, A(t) = 1) \\
 & \quad \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) \\
 & \quad \Pr(T \leq t | \mathbf{X} = \mathbf{x}, A(t) = 1) du \\
 &= \int_0^\infty \phi(u) \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) \\
 & \quad \Pr(T \leq t | \mathbf{X} = \mathbf{x}, A(t) = 1) du \\
 &= p(\mathbf{x}) \left\{ \int_0^{T^*} \phi(u) \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) du + \int_{T^*}^\infty \beta_{T^*} \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) du \right\} \\
 &= p(\mathbf{x}) \left[\int_0^{T^*} \phi(u) \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) du + \beta_{T^*} \right. \\
 & \quad \left. \left\{ 1 - \int_0^{T^*} \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) du \right\} \right]
 \end{aligned}$$

For $s \in [t - T^*, t]$, we write $p_i^*(s | \mathbf{x}) = \Pr(T \leq s | \mathbf{X} = \mathbf{x}, A(t) = 1)$ and $\lambda_i^*(s | \mathbf{x}) = \Pr(T = s | T \geq s, \mathbf{X} = \mathbf{x}, A(t) = 1)$ as the incidence and prevalence of individuals with $\mathbf{X} = \mathbf{x}$ at time s restricted to the eligible population at time t . Clearly, they differ from $p(s | \mathbf{x})$ and $\lambda(s | \mathbf{x})$ since they are restricted to a population that is eligible at a different time. To connect the restricted incidence and prevalence with unrestricted ones, we make an assumption similar to that in Gao and Bannick [3] that $p_i^*(s | \mathbf{x}) = p(s | \mathbf{x}) = p(\mathbf{x})$ and $\lambda_i^*(s | \mathbf{x}) = \lambda(s | \mathbf{x}) = \lambda(\mathbf{x})$. Now, we have

$$\begin{aligned} & \Pr(T = t - u | T \leq t, \mathbf{X} = \mathbf{x}, A(t) = 1) \\ &= \frac{\Pr(T = t - u | \mathbf{X} = \mathbf{x}, A(t) = 1)}{\Pr(T \leq t | \mathbf{X} = \mathbf{x}, A(t) = 1)} \\ &= \frac{\Pr(T = t - u | T \geq t - u, \mathbf{X} = \mathbf{x}, A(t) = 1) \Pr(T \geq t - u | \mathbf{X} = \mathbf{x}, A(t) = 1)}{\Pr(T \leq t | \mathbf{X} = \mathbf{x}, A(t) = 1)} \\ &= \frac{\lambda_i^*(t - u | \mathbf{x}) \{1 - p_i^*(t - u | \mathbf{x})\}}{p(\mathbf{x})} \\ &= \frac{\lambda(\mathbf{x}) \{1 - p(\mathbf{x})\}}{p(\mathbf{x})} \end{aligned}$$

such that

$$\begin{aligned} & \Pr(D = 1, R = 1 | \mathbf{X} = \mathbf{x}, A(t) = 1) \\ &= p(\mathbf{x}) \left[\int_0^{T^*} \phi(u) \frac{\lambda(\mathbf{x}) \{1 - p(\mathbf{x})\}}{p(\mathbf{x})} du + \beta_{T^*} \left\{ 1 - \int_0^{T^*} \frac{\lambda(\mathbf{x}) \{1 - p(\mathbf{x})\}}{p(\mathbf{x})} du \right\} \right] \\ &= p(\mathbf{x}) \left[\Omega_{T^*} \frac{\lambda(\mathbf{x}) \{1 - p(\mathbf{x})\}}{p(\mathbf{x})} + \beta_{T^*} \left\{ 1 - T^* \frac{\lambda(\mathbf{x}) \{1 - p(\mathbf{x})\}}{p(\mathbf{x})} \right\} \right] \\ &= (\Omega_{T^*} - \beta_{T^*} T^*) \lambda(\mathbf{x}) \{1 - p(\mathbf{x})\} + \beta_{T^*} p(\mathbf{x}) \end{aligned}$$

Therefore, the numerator of (1) satisfies

$$\begin{aligned} & \frac{1}{N} \sum_{i=1}^N D_i (R_i - \hat{\beta}_{T^*}) \xrightarrow{P} E\{D_i (R_i - \hat{\beta}_{T^*})\} \\ &= E\{E[D_i (R_i - \hat{\beta}_{T^*}) | \mathbf{X}_i]\} \\ &= E\{E(D_i R_i | \mathbf{X}_i) - E(D_i \hat{\beta}_{T^*} | \mathbf{X}_i)\} \\ &= E\{(\Omega_{T^*} - \beta_{T^*} T^*) \lambda(\mathbf{X}_i) \{1 - p(\mathbf{X}_i)\} + \beta_{T^*} p(\mathbf{X}_i) - \beta_{T^*} p(\mathbf{X}_i)\} \\ &= (\Omega_{T^*} - \beta_{T^*} T^*) E\{\lambda(\mathbf{X}_i) \{1 - p(\mathbf{X}_i)\}\} \end{aligned}$$

where $\xrightarrow{P}$ denotes convergence in probability, and $E(\cdot)$ denotes the expectation with respect to the population with cross-sectional recency testing data, that is, those with $A(t) = 1$.

Similarly, the denominator of (1) satisfies

$$\begin{aligned} & \frac{1}{N} \sum_{i=1}^N (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*) \xrightarrow{P} E\{(1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)\} \\ &= E\{E\{(1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*) | \mathbf{X}_i\}\} \\ &= E\{(\Omega_{T^*} - \beta_{T^*} T^*) E(1 - D_i | \mathbf{X}_i)\} \\ &= (\Omega_{T^*} - \beta_{T^*} T^*) E\{1 - p(\mathbf{X}_i)\} \end{aligned}$$

Then, we get

$$\hat{\lambda} \xrightarrow{P} \frac{E[\lambda(\mathbf{X}_i)\{1 - p(\mathbf{X}_i)\}]}{E\{1 - p(\mathbf{X}_i)\}} = \frac{\int \lambda(\mathbf{x})\{1 - p(\mathbf{x})\}g(\mathbf{x})d\mathbf{x}}{\int \{1 - p(\mathbf{x})\}g(\mathbf{x})d\mathbf{x}}$$

Note that $\{1 - p(\mathbf{x})\}g(\mathbf{x})$ is the density of covariates $\mathbf{X}$ among HIV-negative individuals in the cross-sectional sample. It then implies that the incidence estimator is consistent for the average incidence of HIV-negative individuals in the population. This is in line with modern epidemiological studies on prospective HIV incidence where researchers focus on the HIV incidence estimate of the HIV-seronegative population at baseline [19–21].

2.2.2 | Extension of Cross-Sectional Incidence Estimation to a Target

Population: We consider extending the cross-sectional incidence estimate to a target population that is potentially different from the population with cross-sectional recency testing data, that is, the “cross-sectional population.” Naturally, the target population is composed of HIV-negative individuals only. Let $h(\mathbf{x})$ be the probability density function of covariates $\mathbf{X}$ in the target population. We aim to build a consistent estimator for the average incidence of the target population, that is,

$$\lambda_{\text{target}} \equiv \int \lambda(\mathbf{x})h(\mathbf{x})d\mathbf{x}$$

Motivated by the inverse probability weighting approach, we propose to re-weight the numerator and denominator of $\hat{\lambda}$ to come up with an extended estimator given by

$$\hat{\lambda}_{\text{extend}} = \frac{\sum_{i=1}^N \omega(\mathbf{X}_i) D_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N \omega(\mathbf{X}_i) (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)} \quad (2)$$

Following similar derivations as in Section 2.2.1, the extended estimator follows

$$\begin{aligned} \hat{\lambda}_{\text{extend}} &\xrightarrow{P} \frac{E[\omega(\mathbf{X}_i)\lambda(\mathbf{X}_i)\{1 - p(\mathbf{X}_i)\}]}{E[\omega(\mathbf{X}_i)\{1 - p(\mathbf{X}_i)\}]} \\ &= \frac{\int \omega(\mathbf{x})\lambda(\mathbf{x})\{1 - p(\mathbf{x})\}g(\mathbf{x})d\mathbf{x}}{\int \omega(\mathbf{x})\{1 - p(\mathbf{x})\}g(\mathbf{x})d\mathbf{x}} \end{aligned}$$

Therefore, to make the extended estimator consistent for λ_{target} , the weight should take the form

$$\omega(\mathbf{x}) \propto \frac{h(\mathbf{x})}{\{1 - p(\mathbf{x})\}g(\mathbf{x})} \quad (3)$$

That is, the weight is proportional to the ratio of the densities of covariates $\mathbf{X}$ in the target population and the HIV-negative individuals in the cross-sectional population. Such a weight is intuitive, given that the standard incidence estimator $\hat{\lambda}$ pertains to the average incidence

for the HIV-negative individuals in the cross-sectional population, and we wish to extend such incidence to the target population.

Next, we will discuss how to implement estimator (2) by replacing $\omega(X)$ with its estimator in two different scenarios, where the forms of the estimator differ.

2.2.3 | Weight Estimation When the Target Population is External: We first consider the case when the target population is external to the cross-sectional population. For example, in the design of an HIV prevention trial, researchers may require an estimate of the background incidence for sample size calculation, which can be obtained utilizing the cross-sectional data on recency testing collected in the same geographic area. The distributions of important covariates that affect the incidence and prevalence, that is, $\mathbf{X}$, may be different in the cross-sectional and trial populations such that accounting for population heterogeneity is of interest.

To apply the extended estimator in (2), we need to estimate the weight $\omega(\mathbf{x})$. Specifically, we define a pooled population by combining the cross-sectional population and the target population. For individual $k = 1, \dots, K$ in the pooled population, let S_k indicate which population individual k is from, where $S_k = 1$ indicates the target population and $S_k = 0$ indicates the cross-sectional population. Then, the weight (3) can be written as

$$\begin{aligned}\omega_{\text{ext}}(\mathbf{x}) &\propto \frac{f(\mathbf{x} \mid S_k = 1)}{\Pr(D_k = 0 \mid S_k = 0, \mathbf{X}_k = \mathbf{x})f(\mathbf{x} \mid S_k = 0)} \\ &= \frac{\Pr(S_k = 1 \mid \mathbf{X}_k = \mathbf{x})f(\mathbf{x}) / \Pr(S_k = 1)}{\Pr(D_k = 0 \mid S_k = 0, \mathbf{X}_k = \mathbf{x}) \Pr(S_k = 0 \mid \mathbf{X}_k = \mathbf{x})f(\mathbf{x}) / \Pr(S_k = 0)} \\ &= \frac{\Pr(S_k = 1 \mid \mathbf{X}_k = \mathbf{x})}{\Pr(D_k = 0 \mid S_k = 0, \mathbf{X}_k = \mathbf{x}) \Pr(S_k = 0 \mid \mathbf{X}_k = \mathbf{x})} \times \frac{\Pr(S_k = 0)}{\Pr(S_k = 1)} \\ &\propto \frac{\Pr(S_k = 1 \mid \mathbf{X}_k = \mathbf{x})}{\Pr(S_k = 0, D_k = 0 \mid \mathbf{X}_k = \mathbf{x})} = \frac{\Pr(S_k = 1 \mid \mathbf{X}_k = \mathbf{x}, C_k = 1)}{\Pr(S_k = 0, D_k = 0 \mid \mathbf{X}_k = \mathbf{x}, C_k = 1)}\end{aligned}$$

where $f(\mathbf{x} \mid \cdot)$ denotes the generic notation for the probability density function of covariates $\mathbf{X}$ in a population and $C_k = I(S_k = 1) + I(S_k = 0, D_k = 0)$. The final equality provides insight into weight estimation—specifically, that we can construct a restricted pooled population with individuals in the target population ($S_k = 1$) and HIV-negative individuals in the cross-sectional population ($S_k = 0, D_k = 0$), that is, those with $C_k = 1$. A statistical model can then be applied to estimate the conditional probabilities in the numerator and denominator. For instance, we may assume a logistic regression model fitted to the restricted pooled dataset, where the outcome is membership in the target population ($S_k = 1$) as the outcome and the predictors are $\mathbf{X}_k$. With the estimated weight $\hat{\omega}_{\text{ext}}(\mathbf{x})$, the incidence in the target population can be estimated by

$$\hat{\lambda}_{\text{ext}} = \frac{\sum_{i=1}^N \hat{\omega}_{\text{ext}}(\mathbf{X}_i) D_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N \hat{\omega}_{\text{ext}}(\mathbf{X}_i) (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)}$$

Note that even though the weight is constructed based on HIV-negative individuals (in the target population and the cross-sectional population) only, it is applied to both HIV-negative and HIV-positive individuals in the cross-sectional population.

2.2.4 | Weight Estimation When the Target Population is Internal: Now we consider the case when the target population is a subset of the cross-sectional population, for example, when we collect recency testing data from the screening population, based on which we wish to infer a counterfactual placebo incidence for the trial population, accounting for potential heterogeneity. For individual $i = 1, \dots, N$, we define S_i as the indicator of whether individual i belongs to the target population. The weight (3) can be written as

$$\begin{aligned}\omega_{\text{int}}(\mathbf{x}) &\propto \frac{g(\mathbf{x} \mid D_i = 0, S_i = 1)}{\Pr(D_i = 0 \mid \mathbf{X}_i = \mathbf{x})g(\mathbf{x})} \\ &= \frac{\Pr(D_i = 0, S_i = 1 \mid \mathbf{X}_i = \mathbf{x})g(\mathbf{x}) / \Pr(D_i = 0, S_i = 1)}{\Pr(D_i = 0 \mid \mathbf{X}_i = \mathbf{x})g(\mathbf{x})} \\ &\propto \Pr(S_i = 1 \mid D_i = 0, \mathbf{X}_i = \mathbf{x})\end{aligned}$$

The weight is given by the probability that an HIV-negative individual with covariates $\mathbf{x}$ is included in the target population. To estimate this weight, we fit a statistical model for the distribution of the binary variable ($S_i = 1$ versus $S_i = 0$) conditioning on $D_i = 0$ and $\mathbf{X}_i$. For example, we can fit a logistic regression model using data from the HIV-negative individuals (with $D_i = 0$) in the cross-sectional population, where the outcome is S_i and the predictors are $\mathbf{X}_i$. The incidence in the target population can then be estimated by

$$\hat{\lambda}_{\text{int}} = \frac{\sum_{i=1}^N \hat{\omega}_{\text{int}}(\mathbf{X}_i) D_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N \hat{\omega}_{\text{int}}(\mathbf{X}_i) (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)}$$

where $\hat{\omega}_{\text{int}}(\mathbf{x})$ is an estimate for the weight $\omega_{\text{int}}(\mathbf{x})$.

2.3 | Incorporating HIV Subtypes

In this section, we consider incorporating the covariates that modify the properties of recency tests, for example, HIV subtypes. HIV has multiple distinct subtypes: for example, subtype B is the most prevalent viral strain in North America [22], Europe [23–27], and Australia [28], whereas subtype C predominates in Southern Africa [29, 30]. HIV subtypes modify the properties of recency tests such that the MDRI and FRR for the same recency test are different for individuals with different HIV subtypes. Current and future HIV prevention trials are commonly conducted worldwide and enroll individuals of mixed subtypes. For example, HVTN 704/HPTN 085 enrolls men and transgender people who have sex with men in North America, South America, and Switzerland [31]. It is then crucial to propose a cross-sectional incidence estimator based on recency test results of mixed subtypes. In this section, we consider the scenario where the cross-sectional sample involves J HIV subtypes and modify the estimators $\hat{\lambda}$, $\hat{\lambda}_{\text{ext}}$, and $\hat{\lambda}_{\text{int}}$ to account for HIV subtypes.

2.3.1 | Modifying Cross-Sectional Incidence Estimator Incorporating HIV

Subtypes: To incorporate HIV subtypes in cross-sectional incidence estimation, existing literature proposed to use the weighted MDRI and FRR based on the sample proportions of each subtype [32, 33]. Such an incidence estimator is given by

$$\hat{\lambda}_{\text{original}}^{\text{sub}} = \frac{\sum_{i=1}^N D_i R_i - \sum_{i=1}^N D_i \sum_{j=1}^J \hat{\pi}_j \hat{\beta}_{T^*,j}}{\sum_{i=1}^N (1 - D_i) \left(\sum_{j=1}^J \hat{\pi}_j \hat{\Omega}_{T^*,j} - \sum_{j=1}^J \hat{\pi}_j \hat{\beta}_{T^*,j} \right)}$$

where $\hat{\Omega}_{T^*,j}$ and $\hat{\beta}_{T^*,j}$ are the estimated MDRI and FRR of HIV subtype j , respectively, $\hat{\pi}_j = N_j/N$ is the estimated proportion for subtype j , and N_j is the number of individuals in the cross-sectional sample corresponding to subtype j . As shown in Appendix A, with N and the sample size of the external study for the estimation of $\hat{\Omega}_{T^*,j}$ and $\hat{\beta}_{T^*,j}$ going to infinity,

$$\hat{\lambda}_{\text{original}}^{\text{sub}} \xrightarrow{P} \frac{\sum_{j=1}^J \left\{ (\Omega_{T^*,j} - \beta_{T^*,j}) (1 - p_j) \pi_j \lambda_j + p_j \pi_j \beta_{T^*,j} \right\} - \left(\sum_{j=1}^J p_j \pi_j \right) \left(\sum_{j=1}^J \pi_j \beta_{T^*,j} \right)}{\left(1 - \sum_{j=1}^J p_j \pi_j \right) \left(\sum_{j=1}^J \pi_j (\Omega_{T^*,j} - \beta_{T^*,j}) \right)}$$

where π_j , p_j , λ_j , $\Omega_{T^*,j}$, and $\beta_{T^*,j}$ are the proportion, HIV prevalence, incidence, MDRI, and FRR of subtype j , respectively. It indicates that $\hat{\lambda}_{\text{original}}^{\text{sub}}$ is not consistent for the average incidence in the cross-sectional population.

Let U_i denote the subtype that individual i is associated with for $i = 1, \dots, N$. We propose another estimator

$$\hat{\lambda}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j \frac{\sum_{i=1}^N I(U_i = j) D_i (R_i - \hat{\beta}_{T^*,j})}{\sum_{i=1}^N I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j})}$$

This estimator estimates the incidences associated with each subtype separately and weights them by their sample proportions, which is consistent with the average incidence in the cross-sectional population. Details of proofs appear in Appendix A.

2.3.2 | Accounting for Population Heterogeneity: To further incorporate population heterogeneity when both the cross-sectional and target populations have mixed subtypes, we propose the following estimators:

$$\hat{\lambda}_{\text{ext}}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j^* \frac{\sum_{i=1}^N \hat{\omega}_{\text{ext},j}(X_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*,j})}{\sum_{i=1}^N \hat{\omega}_{\text{ext},j}(X_i) I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j})}$$

and

$$\hat{\lambda}_{\text{int}}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j^* \frac{\sum_{i=1}^N \hat{\omega}_{\text{int},j}(X_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*,j})}{\sum_{i=1}^N \hat{\omega}_{\text{int},j}(X_i) I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j})}$$

where $\hat{\pi}_j^* = M_j/M$, M , and M_j are the sizes of the data from the target population and those in the data from the target population at risk for HIV subtype j , respectively, $\hat{\omega}_{\text{ext},j}$ and $\hat{\omega}_{\text{int},j}$ are estimators for the subtype-specific weights

$$\omega_{\text{ext},j}(\mathbf{x}) = \frac{\Pr(S_i = 1 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)}{\Pr(S_i = 0, D_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)}$$

and

$$\omega_{\text{int},j}(\mathbf{x}) = \Pr(S_i = 1 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j)$$

The proofs for the consistency of $\hat{\lambda}_{\text{ext}}^{\text{sub}}$ and $\hat{\lambda}_{\text{int}}^{\text{sub}}$ can be found in Appendix A.

2.4 | Variance and Confidence Interval Estimation

We propose to estimate the variances of the proposed incidence estimators by a parametric bootstrap approach, since the variances of the estimated $\hat{\beta}_{T^*}$ and $\hat{\Omega}_{T^*}$ are usually reported from the external study. For each bootstrapped sample, we randomly select individuals from the cross-sectional population and the target population with replacement separately and randomly draw the MDRI and FRR from independent log-normal distributions, where the means and variances of the distributions are given by the point estimates and variance estimates, respectively. We apply the proposed methods to the bootstrapped sample to obtain the bootstrap estimates. The variances of the proposed incidence estimators can then be estimated by the empirical variances of bootstrap estimates, and the confidence intervals can be constructed based on the point estimate and the estimated variance of the log-transformed incidence.

When data from the external study are also available, a nonparametric approach may also be used for variance estimation, where the external data are also bootstrapped to estimate β_{T^*} and Ω_{T^*} in each bootstrap round.

3 | Simulation Study

In this section, we will evaluate the numerical performance of the proposed estimators in realistic simulated settings, and compare it with that of naive estimators. In Section 3.1, we describe the general simulation strategy, including how to simulate the cross-sectional sample, estimates for MDRI and FRR to be applied in incidence estimation, and data from a target population. When the data are of the same subtype, we evaluate the performance of the proposed extended estimators $\hat{\lambda}_{\text{ext}}$ and $\hat{\lambda}_{\text{int}}$ in Sections 3.2 and 3.3, respectively. We will also consider the setting where the cross-sectional population (and the target population) is a mixture of individuals at risk for two HIV subtypes. The proposed incidence estimators $\hat{\lambda}^{\text{sub}}$, $\hat{\lambda}_{\text{ext}}^{\text{sub}}$, and $\hat{\lambda}_{\text{int}}^{\text{sub}}$ will be evaluated in Section 3.4.

3.1 | General Simulation Strategy

We consider the simulation setting that mimics a men-who-have-sex-with-men (MSM) population, which is a common population of interest for HIV prevention. Four binary covariates are considered to mimic variables associated with both HIV prevalence and incidence for MSM, including rectal infections [34–36], receptive anal sex [37–39], unprotected anal sex [40, 41], and college education [38, 41]. We specify the HIV prevalence and incidence for individuals with different combinations of those variables based on the average prevalences and incidences and risk ratios reported in the literature, which are given in Supporting Information Table S1.

For an individual who has been infected for s units of time, the recency test result is generated from a Bernoulli distribution with success probability $\phi(s) = 1 - F_{\text{Gamma}}(s; \alpha = 11.40, \beta = 23.66)$, where $F_{\text{Gamma}}(\cdot; \alpha, \beta)$ is the cumulative distribution function of a Gamma random variable with shape α and rate β . This recency test has an MDRI of 176 days and an FRR of 0% with $T^* = 2$, and mimics a commonly applied recency test algorithm (LAg Avidity (Sedia HIV-1 LAg Avidity EIA; Sedia Biosciences Corporation, Portland, OR, USA) ODn < 1.5 and viral load > 1000 copies/mL) for HIV subtype B individuals discussed in Sempa et al. [33].

In each simulation, our data generation consists of the following steps:

- We simulate the cross-sectional data with size N that includes individuals' covariates, HIV status, and recency test results. Specifically, the covariates are generated from a multivariate Bernoulli distribution with parameters given in Supporting Information Table S1. For each combination of covariates, individuals' HIV status at cross-sectional time, (unobserved) HIV infection duration, and recency test results for HIV-infected individuals are simulated based on the procedure discussed in Section 3.3.2 of Gao and Bannick [3] with prevalences and incidences (conditional on covariates) specified in Supporting Information Table S1.
- We simulate the estimates for MDRI and FRR following the procedure in Section 3.3.1 of Gao and Bannick [3]. Specifically, we estimate MDRI based on a simulated dataset comprising 175 individuals and approximately 1000 longitudinal measures of recency test results; we estimate FRR based on another simulated dataset with 1500 individuals who are infected longer than T^* .
- We simulate the target population of size M that includes individuals' covariates only. The procedures are slightly different when the target population is external versus internal and will be described in the following sections.
- We apply the proposed estimators to estimate the incidence in the target population, with cross-sectional data generated in (a), MDRI and FRR estimated in (b), and target population data in (c). We fit logistic regression models to estimate the weights in proposed estimators, where the predictors are the four binary covariates considered in the simulations without any interaction term.

- We estimate the variability of the proposed estimators and build the confidence intervals of interest using the parametric bootstrap approach described in Section 2.4, with 500 bootstrap rounds.

We consider two settings: the first setting with $(N, M) = (2500, 1250)$ and the second one with larger sample sizes $(N, M) = (5000, 2500)$. Based on 500 simulations, we will summarize the results using empirical mean biases, empirical standard errors, mean standard error estimates, and empirical coverage probabilities of 95% confidence intervals, where the “true” incidence is the average incidence among individuals in the target population.

3.2 | External Target Population

We first consider the simulation when the target population is external to the cross-sectional population such that the covariates from the target population and the cross-sectional population are simulated independently and follow the distributions given in Supporting Information Table S1. The true background incidence of the target population is 0.0259 cases per person-year (PY).

We apply the proposed incidence estimator $\hat{\lambda}_{\text{ext}}$ to estimate the incidence for the target population and compare it with the naive incidence estimator $\hat{\lambda}$ that does not account for population heterogeneity. Table 1 shows the simulation results. The proposed estimator $\hat{\lambda}_{\text{ext}}$ has a smaller bias compared to the naive estimator $\hat{\lambda}$. The coverage probability of the 95% confidence interval using the proposed approach is closer to 95%.

3.3 | Internal Target Population

We consider another simulation setting when the target population is part of the cross-sectional population. This mimics the scenario where individuals receive recency testing at the screening stage of an HIV prevention trial, and those who are HIV-negative will be invited to enroll in the trial. We randomly enroll a subset of HIV-negative individuals from the cross-sectional screening population, with the covariate-specific enrollment probabilities provided in Table S1 in the Supporting Information. Specifically, the enrollment probabilities were calculated such that the distribution of the covariates in the trial population is similar to that in the external target population in Section 3.2. The true counterfactual placebo incidence is 0.0248 cases per PY.

In addition to the estimation of incidence in the target population, we also consider the estimation of prevention efficacy when individuals in the trial receive an active prevention product and are followed for HIV infection. For each individual with covariate $\mathbf{x}$ in the active-arm trial, the infection time is simulated from a Poisson process with incidence rate $\lambda(\mathbf{x})(1 - \text{PE})$, where the prevention efficacy $\text{PE} = 0.5$. We assume that each individual is followed for up to 3 years. Let n_{trial} and y_{trial} be the number of HIV infections and person-time in the active-arm trial. The HIV incidence in the trial can be estimated by $\hat{\lambda}_{\text{trial}} = n_{\text{trial}}/y_{\text{trial}}$ and the prevention efficacy of the active product can be estimated by $1 - \hat{\lambda}_{\text{trial}}/\hat{\lambda}$ and $1 - \hat{\lambda}_{\text{trial}}/\hat{\lambda}_{\text{int}}$ based on the naive incidence estimator and the proposed estimator, respectively.

Table 2 presents the simulation results. For the estimation of the incidence in the target population and prevention efficacy of the active prevention product, the proposed estimator shows a smaller bias than the naive estimator. The coverage probability of the 95% confidence interval based on the proposed estimator is closer to the nominal level than the naive estimator.

3.4 | Incorporating HIV Subtypes

Finally, we consider simulation settings when there is a mixture of HIV subtypes, for example, subtypes A and B. The characteristics of recency testing for individuals susceptible to subtype B have been described in Section 3.1. For subtype A, the recency test result is generated from a Bernoulli distribution with success rate $\phi_A(s) = (1 - F_{\text{Gamma}}(s; \alpha = 0.84, \beta = 1.66))I(s \leq 2) + 0.026I(s > 2)$, which matches the MDRI (186 days) and FRR (2.6%) reported in Sempa et al. [33].

We assume that the cross-sectional population includes equal numbers of individuals susceptible to subtypes A and B. In our simulation, the incidence and prevalence corresponding to subtype A are 0.02 cases per PY and 0.25, respectively, and the incidence and prevalence of subtype B are 0.05 cases per PY and 0.15, respectively, which leads to a true average incidence of 0.035 cases per PY. The simulation results for the proposed estimator $\hat{\lambda}^{\text{sub}}$ and the naive estimator $\hat{\lambda}_{\text{original}}^{\text{sub}}$ are given in Table 3. The bias of the proposed estimator is smaller than the naive estimator, and the coverage probability corresponding to the proposed estimator is closer to 95% than the naive estimator.

We also consider the estimation of incidence in a different target population with mixed subtypes. The incidence, prevalence, proportions of subtypes in the cross-sectional and target populations, and the enrollment probabilities are given in Table S2 in the Supporting Information. The true background incidences of the target population are 0.0269 and 0.0252 cases per PY when the target population is external and internal, respectively. Table 4 gives the results comparing the naive estimator $\hat{\lambda}^{\text{sub}}$ and the proposed estimators $\hat{\lambda}_{\text{ext}}^{\text{sub}}$ and $\hat{\lambda}_{\text{int}}^{\text{sub}}$. Similar to the results in Tables 1 and 2, the bias of the proposed estimator is smaller than that of the naive estimator, and the coverage probability of the 95% confidence interval built on the proposed estimator is either comparable or closer to 95% compared with that of the naive estimator.

4 | Data Application

To further demonstrate the application and performance of the proposed methods, we analyze the data from the Sabes Study, which is a multisite study that investigated the feasibility and benefits of incorporating frequent testing for acute and recent HIV infection and rapid linkage to care at community testing and treatment sites among cisgender men who have sex with men and transgender women in Lima, Peru [42]. In step 1 of the study, 3337 high-risk participants were recruited and tested between July 2013 and September 2015. We exclude 228 HIV-uninfected participants who had been recently tested for HIV in the Peruvian Biobehavioral Survey before recruitment and base our analysis on a cross-sectional sample with 3109 participants. Out of the 643 HIV-seropositive individuals, 614

took the recency test, which classifies an individual as recently infected if their Maxim HIV-1 LAg Avidity plasma EIA ODn < 1.5 and viral load > 1000 copies/mL. This recency test has an MDRI of 202 days and an FRR of 0% with $T^* = 2$ for subtype B, which is the most prevalent HIV-1 variant in Latin America [43]. In step 2 of the study, 1912 HIV-negative individuals were enrolled for monthly HIV testing with a 2-year follow-up. Our target population is the step 2 population in the study. We will estimate the HIV incidence of the target population using the recency testing data collected in step 1, and compare it with the prospective follow-up incidence. We consider risk factors that potentially modify HIV prevalence and incidence, including age (18–22, 23–26, 27–32, or > 32), sexuality (homosexual, bisexual, heterosexual, or transgender), sexual role (insertive, receptive, or versatile), postsecondary education (yes or no), number of sex partners (0, 1, 2–5, or > 5), anal sex without condom use (yes or no), anal or vaginal sex with HIV positive partners (yes, no, or do not know), whether an individual has a main partner (yes or no), transactional sex (yes or no), alcohol use (yes or no), and sex under influence (yes or no). To account for the fact that not all HIV-positive individuals took the recency test, we apply modified versions of $\hat{\lambda}$ and $\hat{\lambda}_{\text{int}}$, which are given by

$$\hat{\lambda}^{\text{mod}} = \frac{\sum_{i=1}^N D_i Q_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)} \times \frac{\sum_{i=1}^N D_i}{\sum_{i=1}^N D_i Q_i}$$

and

$$\hat{\lambda}_{\text{int}}^{\text{mod}} = \frac{\sum_{i=1}^N \hat{\omega}_{\text{int}}(X_i) D_i Q_i (R_i - \hat{\beta}_{T^*})}{\sum_{i=1}^N \hat{\omega}_{\text{int}}(X_i) (1 - D_i) (\hat{\Omega}_{T^*} - \hat{\beta}_{T^*} T^*)} \times \frac{\sum_{i=1}^N D_i}{\sum_{i=1}^N D_i Q_i}$$

respectively, where Q_i is the indicator of receiving the recency test for an HIV-positive individual i . We construct the modified estimators based on the assumption that the recency testing is missing at random. Indeed, in the Sabes study, the recency testing is likely to be missing completely at random since individuals (< 5%) failed to be tested by recency testing mainly due to insufficient blood samples and/or inadequate storage of samples. Similar to the simulation studies, we use the parametric bootstrap approach to estimate the confidence interval with 500 bootstrap rounds.

The incidences of the step 2 study population estimated by $\hat{\lambda}^{\text{mod}}$ and $\hat{\lambda}_{\text{int}}^{\text{mod}}$ are 0.108 cases per PY (95% CI: 0.091, 0.130) and 0.110 cases per PY (95% CI: 0.092, 0.132), respectively. Both estimates closely align with the incidence estimate from the longitudinal follow-up data of step 2, which is 0.109 (95% CI: 0.095, 0.123). This is not surprising given the high enrollment rate (78%) of HIV-negative individuals in step 2. In addition, we found that the risk factors were distributed similarly among the HIV-negative individuals in step 1 and step 2, which is reflected by the distribution of the estimated weight $\hat{\omega}_{\text{int}}$ (see Figure S1 in the Supporting Information). Specifically, the distribution of the estimated weight standardized by the enrollment rate is concentrated in the range of 0.9–1.2, indicating that individuals were enrolled in the step 2 study in a “uniform” manner.

To further illustrate the proposed methods, we consider the setting: when the target population includes subpopulations of the Sabes step 2 study. This mimics the scenarios when the trial aims to enroll high-risk individuals among those screened. Specifically, we consider the subpopulations that include (1) individuals of age ≤ 32 , and (2) individuals with at least 1 sex partner. The incidences of the target populations estimated by $\hat{\lambda}^{\text{mod}}$, $\hat{\lambda}_{\text{int}}^{\text{mod}}$, and prospective follow-up data are shown in Table 5. For both sub-populations, the proposed estimator $\hat{\lambda}_{\text{int}}^{\text{mod}}$ outperforms the naive estimator $\hat{\lambda}^{\text{mod}}$ in that the estimate is closer to the longitudinal incidence estimate.

5 | Discussion

In this paper, we outlined a statistical framework to estimate the incidence of a different target population addressing population heterogeneity for HIV incidence estimation based on the recency test. We considered two specific settings when the target population is a subset of the cross-sectional population and when the target population is external. In addition, we proposed estimators to incorporate HIV subtypes that modify the properties of recency tests. The properties of the proposed estimators were illustrated in simulation studies and a data application.

The proposed estimators require that the covariates $\mathbf{X}$ include all the factors that modify HIV prevalence and incidence. Such an assumption may need careful examination, especially given that here $\mathbf{X}$ is assumed to be time-independent. Indeed, there may be time-dependent variables, for example, recent sexual behaviors, that also modify HIV prevalence and incidence. However, it is very difficult to incorporate time-dependent covariates in the proposed framework, as no longitudinal trajectory of the time-dependent covariates can be obtained from individuals in the cross-sectional population. Further research should consider conducting sensitivity analyses to evaluate the performance of the estimators when the assumption is not met. For example, one can perform simulation studies to assess these effects empirically, incorporating time-dependent variables that modify HIV prevalence and incidence in the data generation process.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability Statement

Simulations can be reproduced using the R code at <https://github.com/qii-wang/HIV-incidence-recency-heterogeneity>. The Sabes study data that support the analysis in this paper are from the Sabes study team. The data are available on request from Dr. Ann Duerr (aduerr@fredhutch.org), with the permission of the Sabes study team. The data are not publicly available due to privacy restrictions.

Appendix A: Additional Details of Proofs for HIV Subtypes

Proofs for $\hat{\lambda}_{\text{original}}^{\text{sub}}$ and $\hat{\lambda}^{\text{sub}}$ in Section 2.3.1

Recall that U_i is the HIV subtype of individual i . It can be shown that as N tends to infinity and when $(\hat{\Omega}_{T^*,j}, \hat{\beta}_{T^*,j})$ is consistent for $(\Omega_{T^*,j}, \beta_{T^*,j})$,

$$\begin{aligned} N^{-1} \sum_{i=1}^N D_i R_i &\xrightarrow{P} E(D_i R_i) = \sum_{j=1}^J E(D_i R_i | U_i = j) \Pr(U_i = j) \\ &= \sum_{j=1}^J \{(\Omega_{T^*,j} - \beta_{T^*,j} T^*)(1 - p_j) \lambda_j + p_j \beta_{T^*,j}\} \pi_j, \\ N^{-1} \sum_{i=1}^N D_i &\xrightarrow{P} E(D_i) = \sum_{j=1}^J E(D_i | U_i = j) \Pr(U_i = j) = \sum_{j=1}^J p_j \pi_j \end{aligned}$$

and

$$N^{-1} \sum_{i=1}^N (1 - D_i) \xrightarrow{P} E(1 - D_i) = 1 - \sum_{j=1}^J p_j \pi_j$$

such that the original estimator

$$\begin{aligned} \hat{\lambda}_{\text{original}}^{\text{sub}} &\xrightarrow{P} \frac{\sum_{j=1}^J \{(\Omega_{T^*,j} - \beta_{T^*,j} T^*)(1 - p_j) \pi_j \lambda_j + p_j \pi_j \beta_{T^*,j}\} - \left(\sum_{j=1}^J p_j \pi_j\right) \left(\sum_{j=1}^J \pi_j \beta_{T^*,j}\right)}{\left(1 - \sum_{j=1}^J p_j \pi_j\right) \left\{\sum_{j=1}^J \pi_j (\Omega_{T^*,j} - \beta_{T^*,j} T^*)\right\}} \\ &\neq \sum_{j=1}^J \pi_j \lambda_j \end{aligned}$$

This indicates that $\hat{\lambda}_{\text{original}}^{\text{sub}}$ is inconsistent for the average incidence in the cross-sectional population.

Let

$$A_j = \frac{\sum_{i=1}^N I(U_i = j) D_i (R_i - \hat{\beta}_{T^*,j})}{\sum_{i=1}^N I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j} T^*)}$$

such that $\hat{\lambda}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j A_j$. Note that as N tends to infinity and when $(\hat{\Omega}_{T^*,j}, \hat{\beta}_{T^*,j})$ is consistent for $(\Omega_{T^*,j}, \beta_{T^*,j})$,

$$\begin{aligned} & N^{-1} \sum_{i=1}^N D_i (R_i - \hat{\beta}_{T^*,j}) I(U_i = j) \\ & \xrightarrow{p} E\{D_i (R_i - \hat{\beta}_{T^*,j}) I(U_i = j)\} \\ & = E\{D_i R_i I(U_i = j)\} - E\{D_i \hat{\beta}_{T^*,j} I(U_i = j)\} \\ & = E(D_i R_i | U_i = j) \pi_j - E(D_i | U_i = j) \pi_j \beta_{T^*,j} \\ & = \{(\Omega_{T^*,j} - \beta_{T^*,j}^{T^*})(1 - p_j) \lambda_j + p_j \beta_{T^*,j}\} \pi_j - p_j \pi_j \beta_{T^*,j} \\ & = (\Omega_{T^*,j} - \beta_{T^*,j}^{T^*})(1 - p_j) \lambda_j \pi_j \end{aligned}$$

and

$$N^{-1} \sum_{i=1}^N (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j}^{T^*}) I(U_i = j) \xrightarrow{p} (\Omega_{T^*,j} - \beta_{T^*,j}^{T^*})(1 - p_j) \pi_j$$

Therefore, we get

$$\hat{\lambda}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j A_j \xrightarrow{p} \sum_{j=1}^J \pi_j \lambda_j$$

which implies that $\hat{\lambda}^{\text{sub}}$ is consistent for the average incidence in the cross-sectional population.

Proofs for $\hat{\lambda}_{\text{ext}}^{\text{sub}}$ and $\hat{\lambda}_{\text{int}}^{\text{sub}}$ in Section 2.3.2

Let $p(\mathbf{x}, j)$ and $\lambda(\mathbf{x}, j)$ denote the prevalence and incidence, respectively, for an individual who is eligible at the time of the cross-sectional sample, with covariates $\mathbf{X} = \mathbf{x}$ and subtype j . Let $g(\mathbf{x}, j)$ and $h(\mathbf{x}, j)$ be the joint probability density function of covariates and subtype in the cross-sectional population and target population, respectively. Consider the generic estimator

$$\hat{\lambda}_{\text{extend}}^{\text{sub}} = \sum_{j=1}^J \hat{\pi}_j^* \frac{\sum_{i=1}^N \omega_j(\mathbf{X}_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*,j})}{\sum_{i=1}^N \omega_j(\mathbf{X}_i) I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*,j} - \hat{\beta}_{T^*,j}^{T^*})}$$

where $\omega_j(\cdot)$ is a weight corresponding to subtype j . Note that as N tends to infinity and when $(\hat{\Omega}_{T^*,j}, \hat{\beta}_{T^*,j})$ is consistent for $(\Omega_{T^*,j}, \beta_{T^*,j})$,

$$\begin{aligned}
& N^{-1} \sum_{i=1}^N \omega_j(\mathbf{X}_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*, j}) \\
& \xrightarrow{P} E\left\{ \omega_j(\mathbf{X}_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*, j}) \right\} \\
& = E\left\{ E\left\{ \omega_j(\mathbf{X}_i) I(U_i = j) D_i (R_i - \hat{\beta}_{T^*, j}) \mid \mathbf{X}_i \right\} \right\} \\
& = E\left\{ \omega_j(\mathbf{X}_i) E\left\{ I(U_i = j) D_i (R_i - \hat{\beta}_{T^*, j}) \mid \mathbf{X}_i \right\} \right\} \\
& = E\left\{ \omega_j(\mathbf{X}_i) \left[E\{I(U_i = j) D_i R_i \mid \mathbf{X}_i\} - E\{I(U_i = j) D_i \hat{\beta}_{T^*, j} \mid \mathbf{X}_i\} \right] \right\} \\
& = E\left\{ \omega_j(\mathbf{X}_i) \left[E\{I(U_i = j), D_i, R_i \mid \mathbf{X}_i\} - \beta_{T^*, j} E\{I(U_i = j), D_i \mid \mathbf{X}_i\} \right] \right\} \\
& = E\left\{ \omega_j(\mathbf{X}_i) \left\{ E(D_i, R_i \mid U_i = j, \mathbf{X}_i) - \beta_{T^*, j} E(D_i \mid U_i = j, \mathbf{X}_i) \right\} \Pr(U_i = j \mid \mathbf{X}_i) \right\} \\
& = E\left\{ \omega_j(\mathbf{X}_i) \left[(\Omega_{T^*, j} - \beta_{T^*, j} T^*) \lambda(\mathbf{X}_i, j) \{1 - p(\mathbf{X}_i, j)\} + \beta_{T^*, j} p(\mathbf{X}_i, j) - \beta_{T^*, j} p(\mathbf{X}_i, j) \right] \Pr(U_i = j \mid \mathbf{X}_i) \right\} \\
& = (\Omega_{T^*, j} - \beta_{T^*, j} T^*) E\left\{ \omega_j(\mathbf{X}_i) \lambda(\mathbf{X}_i, j) \{1 - p(\mathbf{X}_i, j)\} \Pr(U_i = j \mid \mathbf{X}_i) \right\}
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
& N^{-1} \sum_{i=1}^N \omega_j(\mathbf{X}_i) I(U_i = j) (1 - D_i) (\hat{\Omega}_{T^*, j} - \hat{\beta}_{T^*, j} T^*) \\
& \xrightarrow{P} (\Omega_{T^*, j} - \beta_{T^*, j} T^*) E\left\{ \omega_j(\mathbf{X}_i) \{1 - p(\mathbf{X}_i, j)\} \Pr(U_i = j \mid \mathbf{X}_i) \right\}
\end{aligned}$$

Furthermore, $\hat{\pi}_j^*$ converges in probability to the true proportion of subtype j in the target population, given by $\int h(\mathbf{x}, j) d\mathbf{x}$. Therefore, if we adopt a weight of the form

$\omega_j(\mathbf{x}) \propto \frac{h(\mathbf{x}, j)}{\{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j)}$, we have

$$\begin{aligned}
\hat{\lambda}_{\text{extend}}^{\text{sub}} & \xrightarrow{P} \sum_{j=1}^J \frac{E\left\{ \omega_j(\mathbf{X}_i) \lambda(\mathbf{X}_i, j) \{1 - p(\mathbf{X}_i, j)\} \Pr(U_i = j \mid \mathbf{X}_i) \right\}}{E\left\{ \omega_j(\mathbf{X}_i) \{1 - p(\mathbf{X}_i, j)\} \Pr(U_i = j \mid \mathbf{X}_i) \right\}} \int h(\mathbf{x}, j) d\mathbf{x} \\
& = \sum_{j=1}^J \frac{\int \omega_j(\mathbf{x}) \lambda(\mathbf{x}, j) \{1 - p(\mathbf{x}, j)\} \Pr(U_i = j \mid \mathbf{x}) g(\mathbf{x}) d\mathbf{x}}{\int \omega_j(\mathbf{x}) \{1 - p(\mathbf{x}, j)\} \Pr(U_i = j \mid \mathbf{x}) g(\mathbf{x}) d\mathbf{x}} \int h(\mathbf{x}, j) d\mathbf{x} \\
& = \sum_{j=1}^J \frac{\int \omega_j(\mathbf{x}) \lambda(\mathbf{x}, j) \{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j) d\mathbf{x}}{\int \omega_j(\mathbf{x}) \{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j) d\mathbf{x}} \int h(\mathbf{x}, j) d\mathbf{x} \\
& = \sum_{j=1}^J \frac{\int \frac{h(\mathbf{x}, j)}{\{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j)} \lambda(\mathbf{x}, j) \{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j) d\mathbf{x}}{\int \frac{h(\mathbf{x}, j)}{\{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j)} \{1 - p(\mathbf{x}, j)\} g(\mathbf{x}, j) d\mathbf{x}} \int h(\mathbf{x}, j) d\mathbf{x} \\
& = \sum_{j=1}^J \frac{\int \lambda(\mathbf{x}, j) h(\mathbf{x}, j) d\mathbf{x}}{\int h(\mathbf{x}, j) d\mathbf{x}} \int h(\mathbf{x}, j) d\mathbf{x} \\
& = \sum_{j=1}^J \int \lambda(\mathbf{x}, j) h(\mathbf{x}, j) d\mathbf{x}
\end{aligned}$$

This implies that $\hat{\lambda}_{\text{extend}}^{\text{sub}}$ is consistent for the average incidence in the cross-sectional population.

Regarding the weight estimation, when the target population is external,

$$\begin{aligned}
\omega_{\text{ext}, j}(\mathbf{x}) &\propto \frac{h(\mathbf{x}, j)}{\{1 - p(\mathbf{x}, j)\}g(\mathbf{x}, j)} \\
&= \frac{f(\mathbf{x}, j \mid S_i = 1)}{\Pr(D_i = 0 \mid S_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j)f(\mathbf{x}, j \mid S_i = 0)} \\
&= \frac{\Pr(S_i = 1 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)f(\mathbf{x}, j) / \Pr(S_i = 1)}{\Pr(D_i = 0 \mid S_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j)} \\
&= \frac{\Pr(S_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)f(\mathbf{x}, j) / \Pr(S_i = 0)}{\Pr(S_i = 1 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)} \\
&= \frac{\Pr(D_i = 0 \mid S_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j) \Pr(S_i = 1)}{\Pr(S_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)} \\
&\propto \frac{\Pr(S_i = 1 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)}{\Pr(S_i = 0, D_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)}
\end{aligned}$$

where $f(\mathbf{x}, j \mid \cdot)$ is the generic notation for the joint probability density function of covariates and subtype in a population. Similarly, when the target population is internal,

$$\begin{aligned}
\omega_{\text{int}, j}(\mathbf{x}) &\propto \frac{h(\mathbf{x}, j)}{\{1 - p(\mathbf{x}, j)\}g(\mathbf{x}, j)} \\
&= \frac{g(\mathbf{x}, j \mid D_i = 0, S_i = 1)}{\Pr(D_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)g(\mathbf{x}, j)} \\
&= \frac{g(\mathbf{x}, j) / \Pr(D_i = 0, S_i = 1)}{\Pr(D_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)g(\mathbf{x}, j)} \\
&= \frac{\Pr(D_i = 0, S_i = 1 \mid \mathbf{X}_i = \mathbf{x}, U_i = j) / \Pr(D_i = 0, S_i = 1)}{\Pr(D_i = 0 \mid \mathbf{X}_i = \mathbf{x}, U_i = j)} \\
&= \frac{\Pr(S_i = 1 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j)}{\Pr(D_i = 0, S_i = 1)} \\
&\propto \Pr(S_i = 1 \mid D_i = 0, \mathbf{X}_i = \mathbf{x}, U_i = j)
\end{aligned}$$

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TABLE 1 |

Simulation results when the target population is external.

Sample size (N, M)	Estimator	Bias $\times 100$	SE $\times 100$	SEE $\times 100$	Cov
(2500, 1250)	Naive	-0.35	0.58	0.48	91.8
	Proposed	0.03	0.70	0.59	94.8
(5000, 2500)	Naive	-0.35	0.48	0.35	83.0
	Proposed	0.02	0.58	0.43	92.2

Note: For each sample size configuration, we report the empirical mean bias (Bias), empirical standard error (SE), mean standard error estimate (SEE), and empirical coverage probability of the 95% confidence interval (Cov). Note that the unit for incidence is cases per PY.

TABLE 2 |
Simulation results when the target population is internal.

Sample size (N, M)	Estimator	Bias $\times 100$	SE $\times 100$	SEE $\times 100$	Cov
Incidence estimation					
(2500, 1250)	Naive	-0.21	0.59	0.49	93.2
	Proposed	0.06	0.68	0.56	93.6
(5000, 2500)	Naive	-0.22	0.52	0.35	88.2
	Proposed	0.05	0.58	0.40	93.4
Prevention efficacy estimation					
(2500, 1250)	Naive	-7.47	17.83	18.18	94.2
	Proposed	-1.27	15.62	16.53	93.6
(5000, 2500)	Naive	-5.80	11.69	11.36	92.4
	Proposed	0.08	10.54	10.34	92.8

Note: For each sample size configuration and simulation scenario, we report the empirical mean bias (Bias), empirical standard error (SE), mean standard error estimate (SEE), and empirical coverage probability of the 95% confidence interval (Cov). Note that the unit for incidence is cases per PY.

TABLE 3 |

Simulation results for incidence estimation in cross-sectional population with mixed subtypes.

Sample size N	Estimator	Bias $\times 100$	SE $\times 100$	SEE $\times 100$	Cov
2500	Naive	0.37	0.72	0.71	90.4
	Proposed	0.05	0.75	0.72	96.6
5000	Naive	0.33	0.53	0.51	88.2
	Proposed	0.01	0.58	0.52	94.4

Note: For each sample size configuration, we report the empirical mean bias (Bias), empirical standard error (SE), mean standard error estimate (SEE), and empirical coverage probability of the 95% confidence interval (Cov). Note that the unit for incidence is cases per PY.

TABLE 4 |

Simulation results for two heterogeneous populations incorporating HIV subtypes.

Sample size (N, M)	Estimator	Bias $\times 100$	SE $\times 100$	SEE $\times 100$	Cov
External target population – incidence estimation					
(2500, 1250)	Naive	-0.37	0.59	0.55	95.4
	Proposed	-0.01	0.73	0.67	96.0
(5000, 2500)	Naive	-0.36	0.44	0.39	88.0
	Proposed	0.00	0.52	0.47	95.2
Internal target population – incidence estimation					
(2500, 1250)	Naive	-0.16	0.61	0.56	96.8
	Proposed	0.01	0.65	0.61	96.2
(5000, 2500)	Naive	-0.16	0.49	0.40	94.4
	Proposed	0.01	0.52	0.44	95.6
Internal target population – prevention efficacy estimation					
(2500, 1250)	Naive	-6.44	18.99	26.46	95.0
	Proposed	-2.62	17.67	29.37	95.6
(5000, 2500)	Naive	-4.33	11.73	12.27	94.6
	Proposed	-0.79	11.08	11.61	95.2

Note: For each sample size configuration and simulation scenario, we report the empirical mean bias (Bias), empirical standard error (SE), mean standard error estimate (SEE), and empirical coverage probability of the 95% confidence interval (Cov). Note that the unit for incidence is cases per PY.

TABLE 5 |

Incidence estimation results with sub-populations of the Sabes step 2 study.

Estimator	Estimate	95% Confidence interval
Individuals of age ≤ 32		
Longitudinal	0.132	(0.114, 0.150)
Naive	0.122	(0.101, 0.147)
Proposed	0.126	(0.104, 0.152)
Individuals with at least 1 sex partner		
Longitudinal	0.110	(0.095, 0.124)
Naive	0.107	(0.089, 0.128)
Proposed	0.109	(0.091, 0.131)

Note: For each estimator across both sub-populations, we report the incidence estimate and 95% confidence interval. Note that the unit for incidence is cases per PY.